

nb-bench-v3

2026-06-15

results from more extensive simulations and maybe real data

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr    1.6.0
## v ggplot2    4.0.1      v tibble     3.3.0
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.2.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

0. introduce my new simulations

main problem: no single nice distribution looks anything like the NPs

The point of this plot: to show how we need the complexity of the summed process.

so what am i doing now? inspired by Cellbender model:

$$Y_i = \text{Pois}(L_i \mu_{\text{exo}}) + \text{Ber}(\pi_{\text{endo}}) \text{Pois}(L_i \mu_{\text{endo}}) + \text{Ber}(\pi_{\text{pert}}) \text{NB}(L_i \mu_{\text{pert}}, \theta_{\text{pert}})$$

The idea:

- the perturbed distribution is the same: NegBin with cell-specific library size rescaling factors
- but now the NP is a sum of two processes:
 - exogeneous noise, representing random stray bits getting in there. μ_{exo} is small
 - endogenous noise, from droplet contamination, chimeras, etc, giving rare but larger values. π_{endo} is very small but μ_{endo} is larger.

I am able to get satisfying looking distributions from this, and in particular I can now simulate distributions where precision is the hard part. Before, with such light tails, precision was often trivial because the NPs were so concentrated near zero.

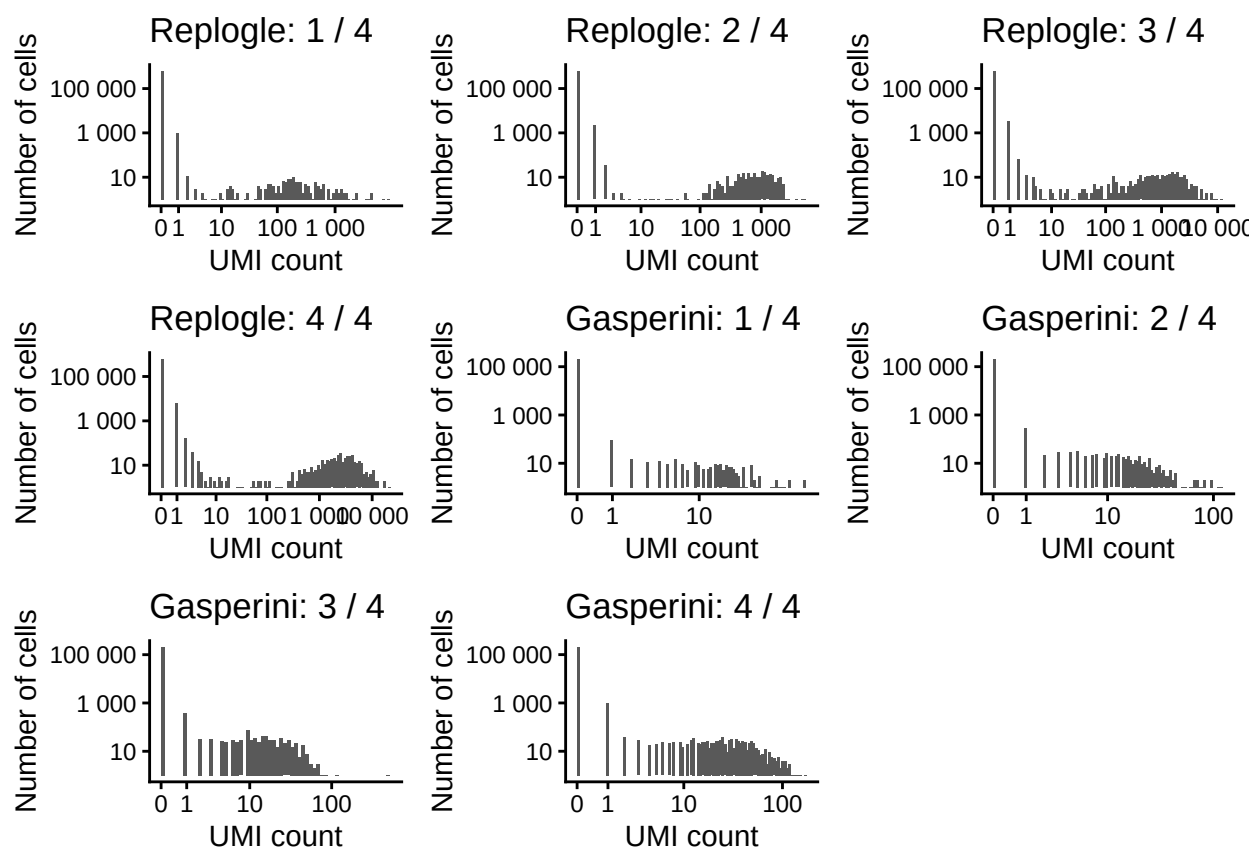
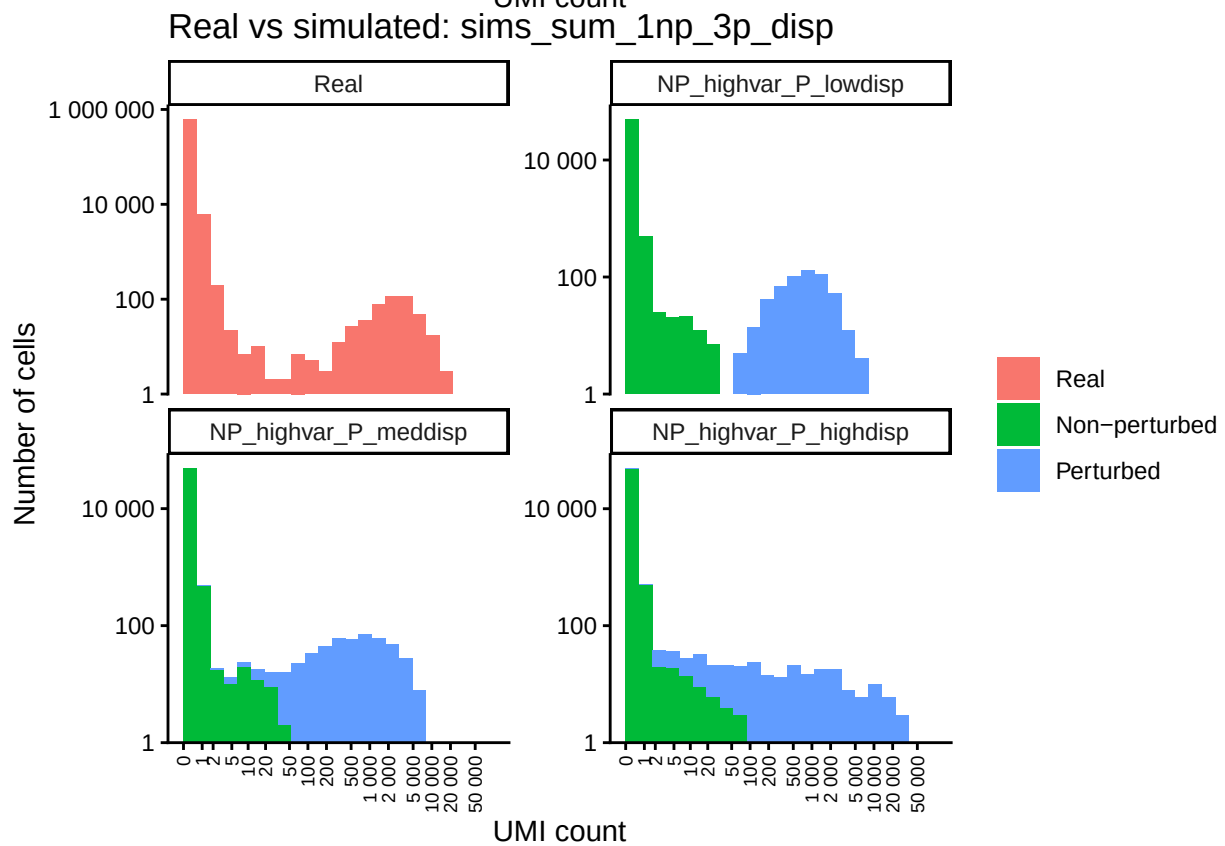
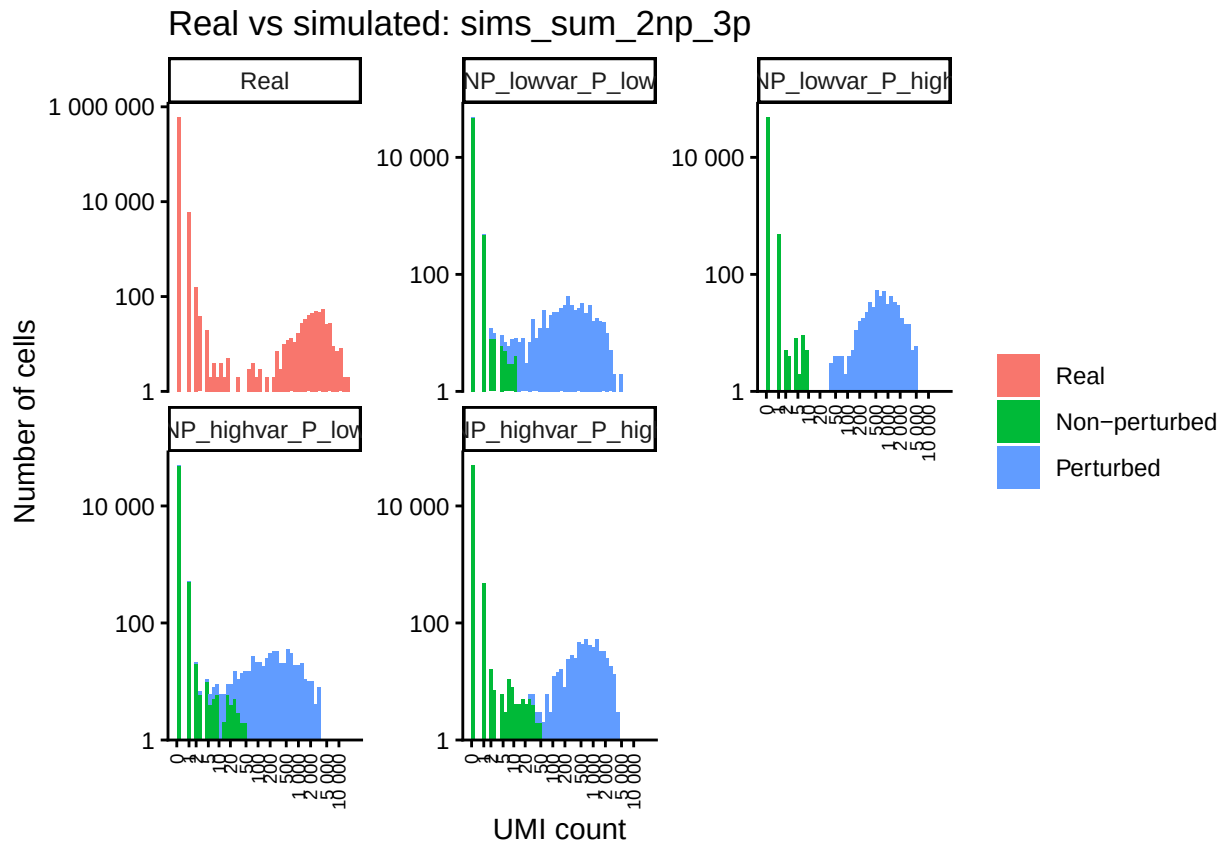
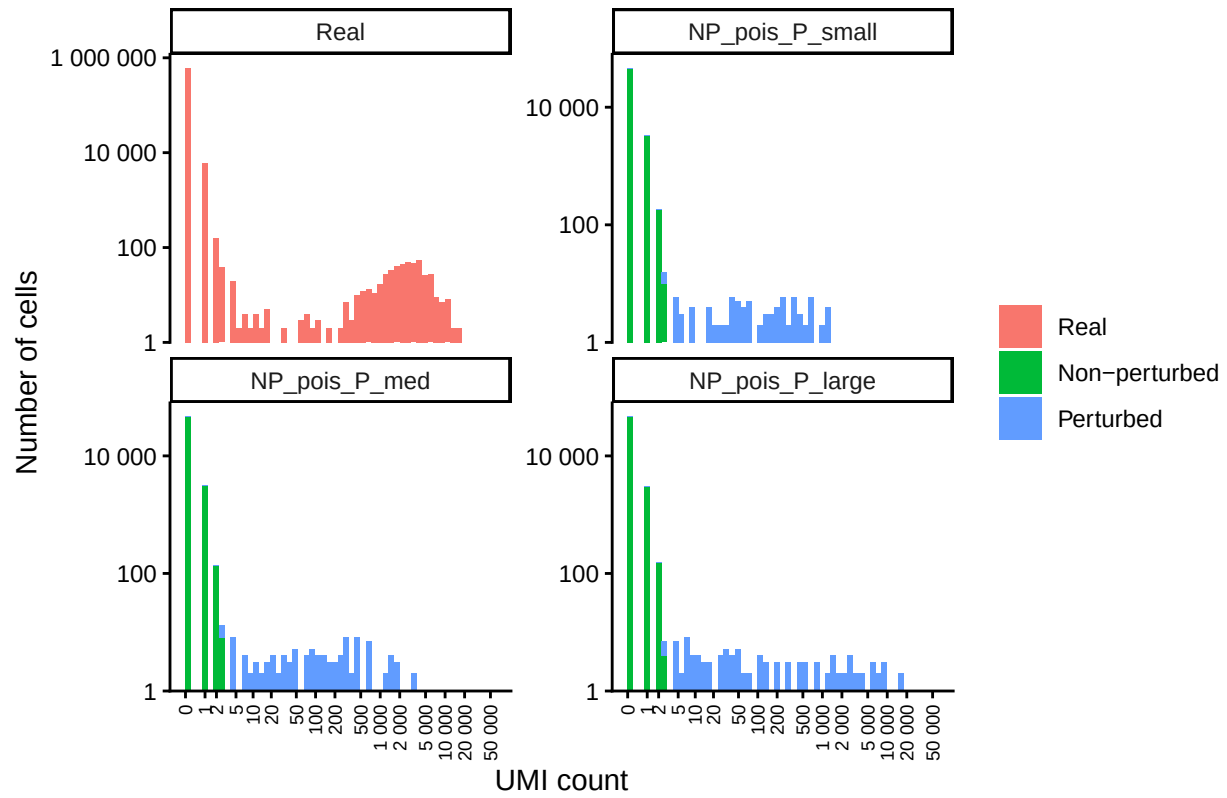


Figure 1: Standard distributions do not decay so rapidly but then become nearly uniform. A single Pois or NB will not provide a realistic non-perturbed model.



Real vs simulated: sims_repeat_old



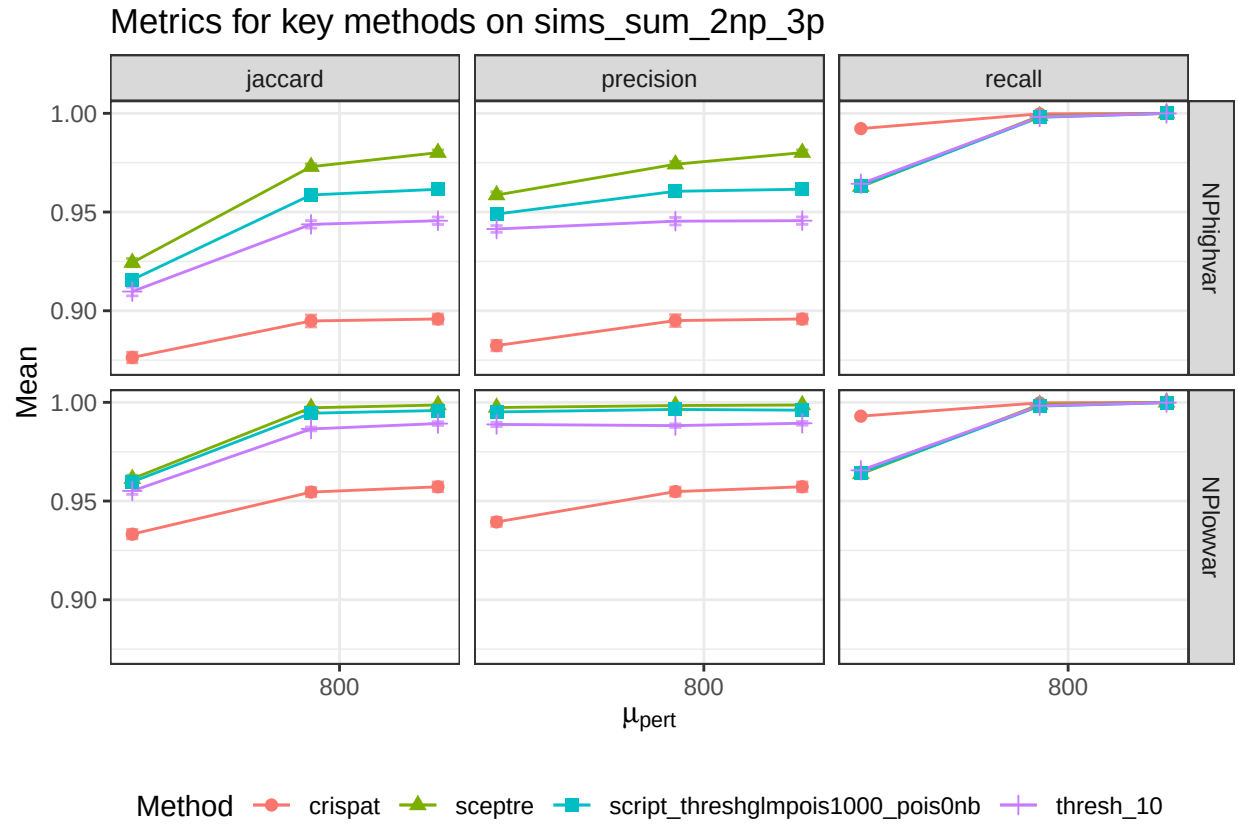


Figure 2: Comparing sceptre, crispat, and the bset new method I've found, on the new simulations sims_sum_2np_3p

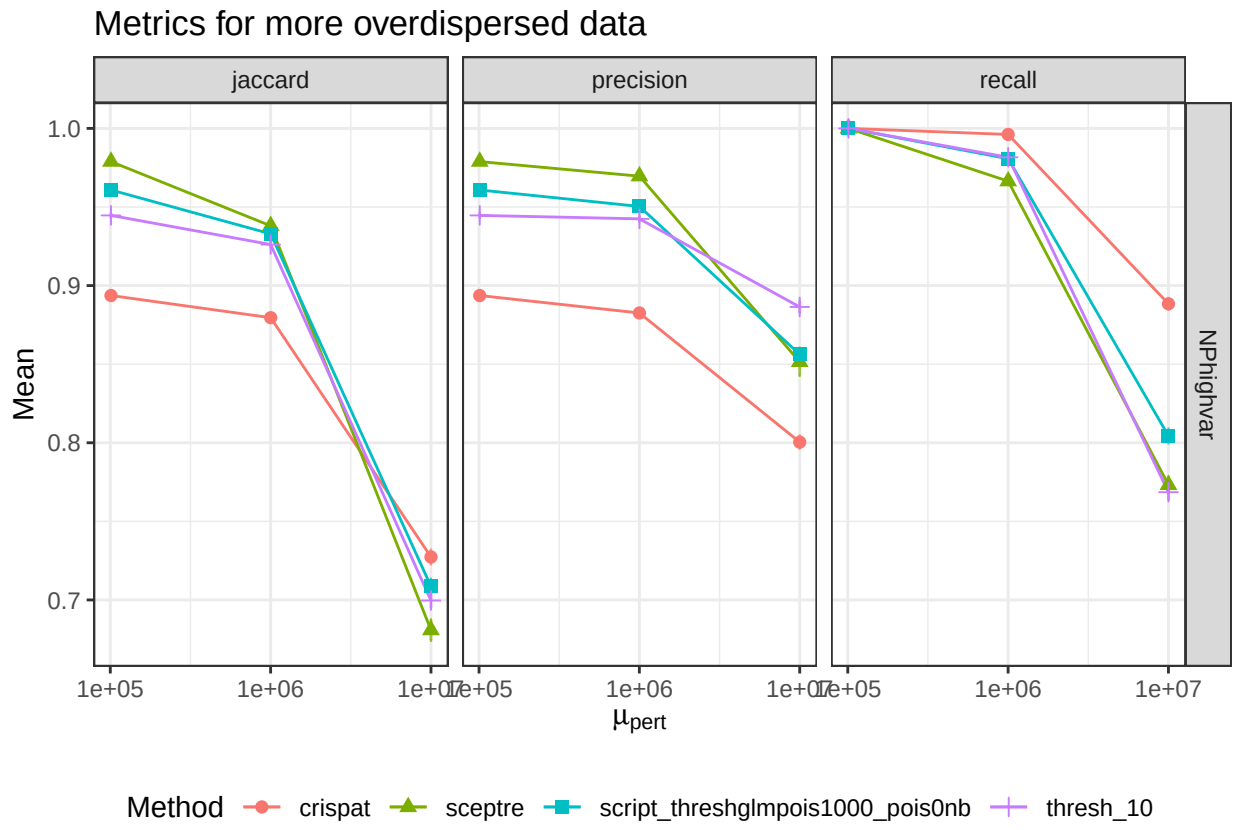


Figure 3: simulations with much bigger and more overdispersed values in sims_sum_1np_3p_disp

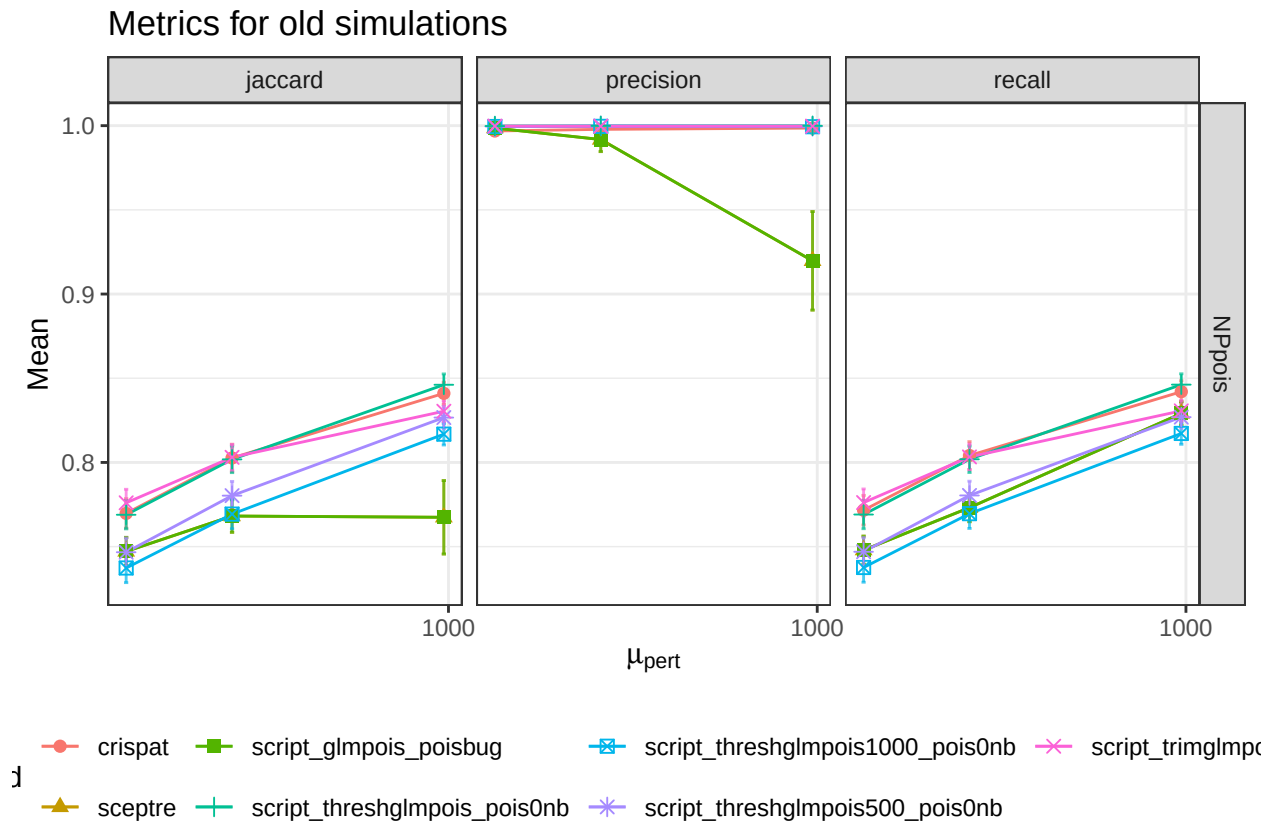


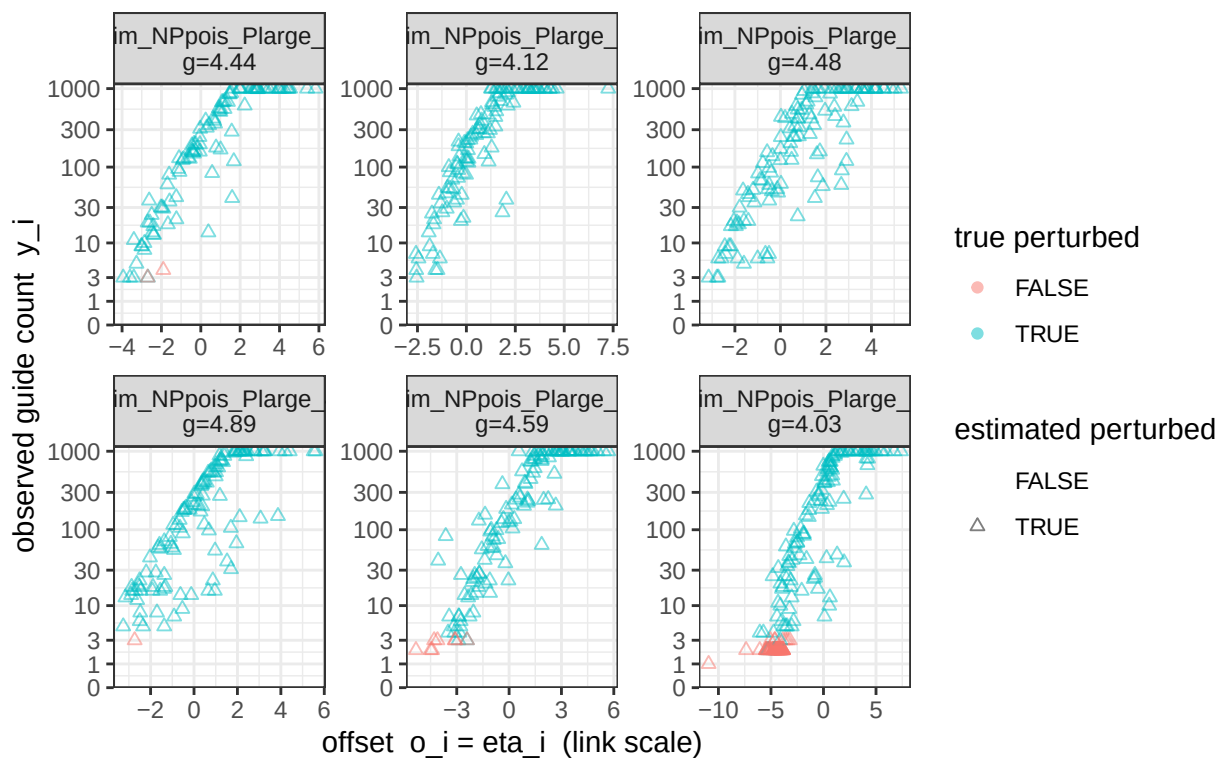
Figure 4: repeating old simulations

1. summary metrics

1.5 why does bugged work so well?

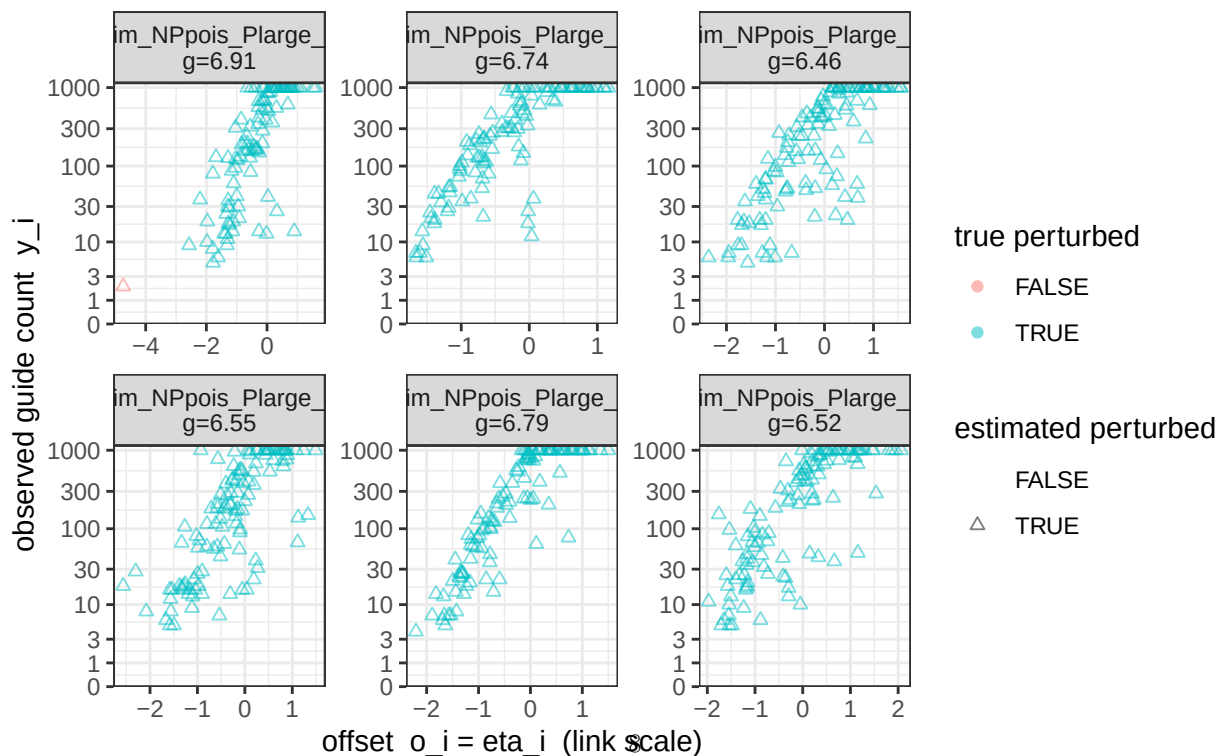
y vs offset, coloured by true perturbed (pois(bugged))

6 guides; up to 10000 cells per panel

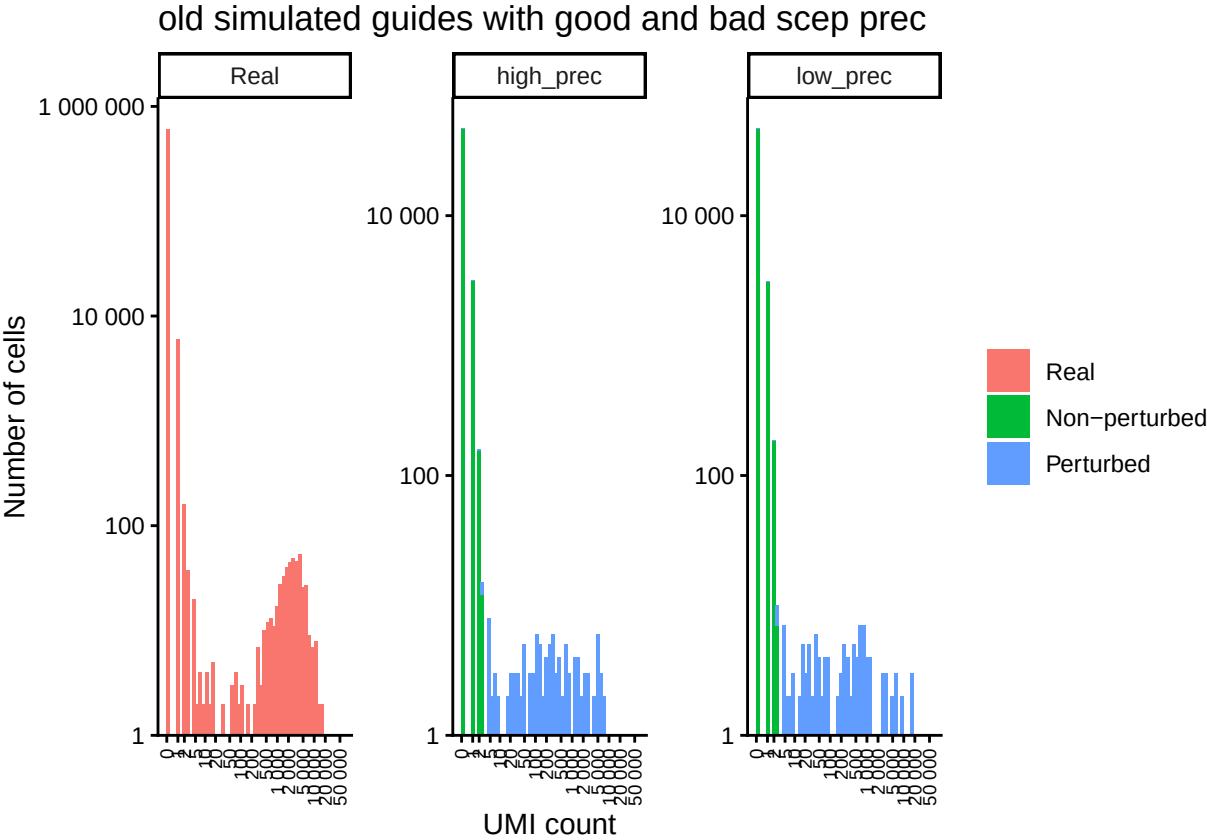


y vs offset, coloured by true perturbed (pois0-nb)

6 guides; up to 10000 cells per panel

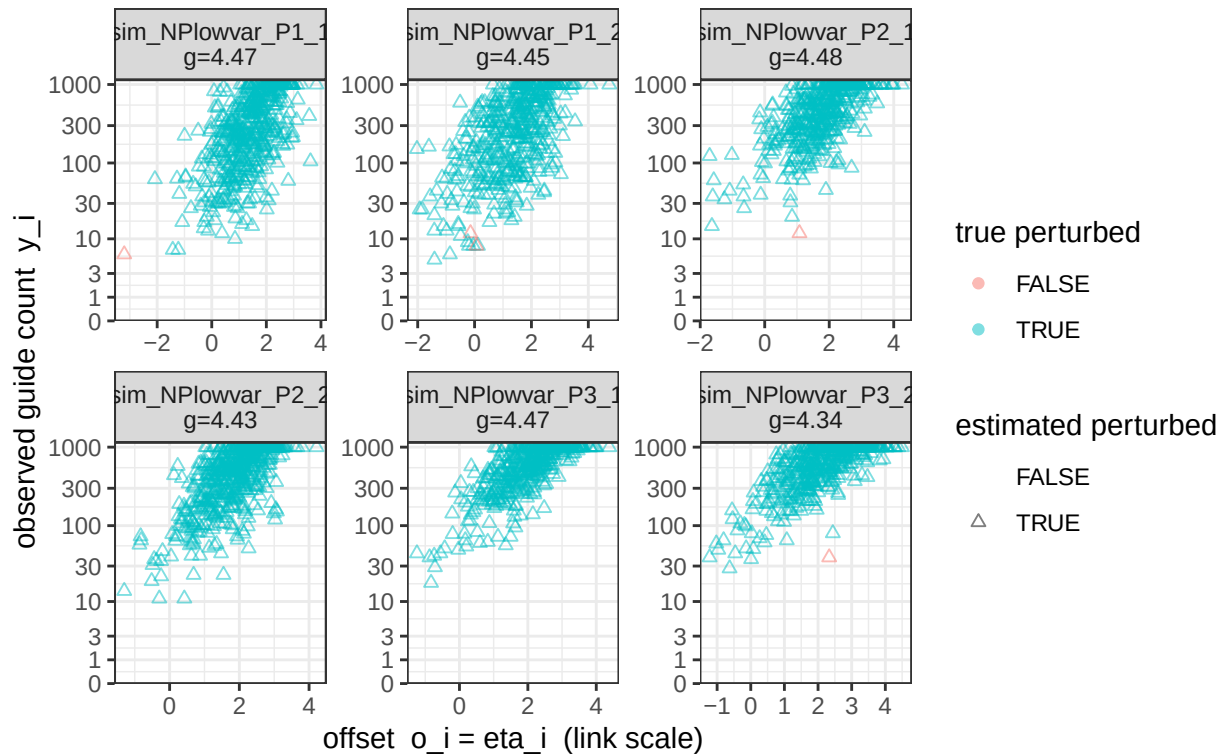


some guides sceptre failed on:



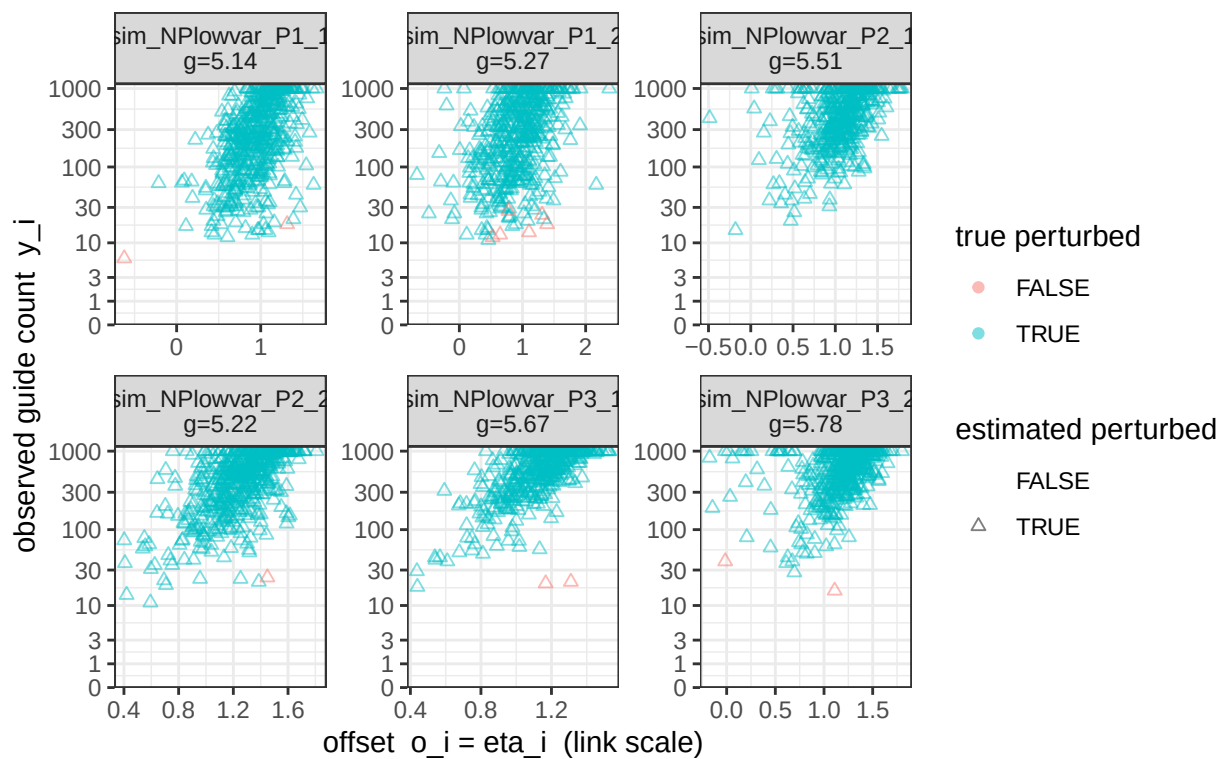
y vs offset, coloured by true perturbed (pois(bugged))

6 guides; up to 10000 cells per panel



y vs offset, coloured by true perturbed (pois0-nb)

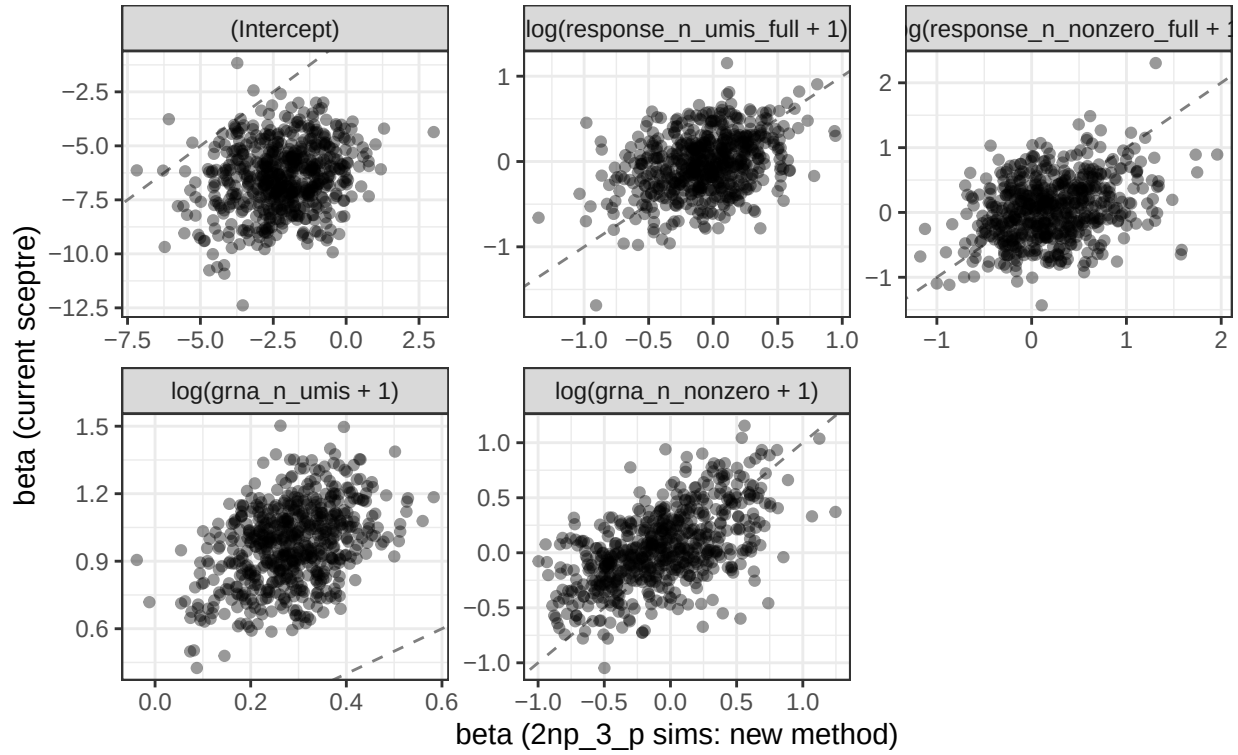
6 guides; up to 10000 cells per panel



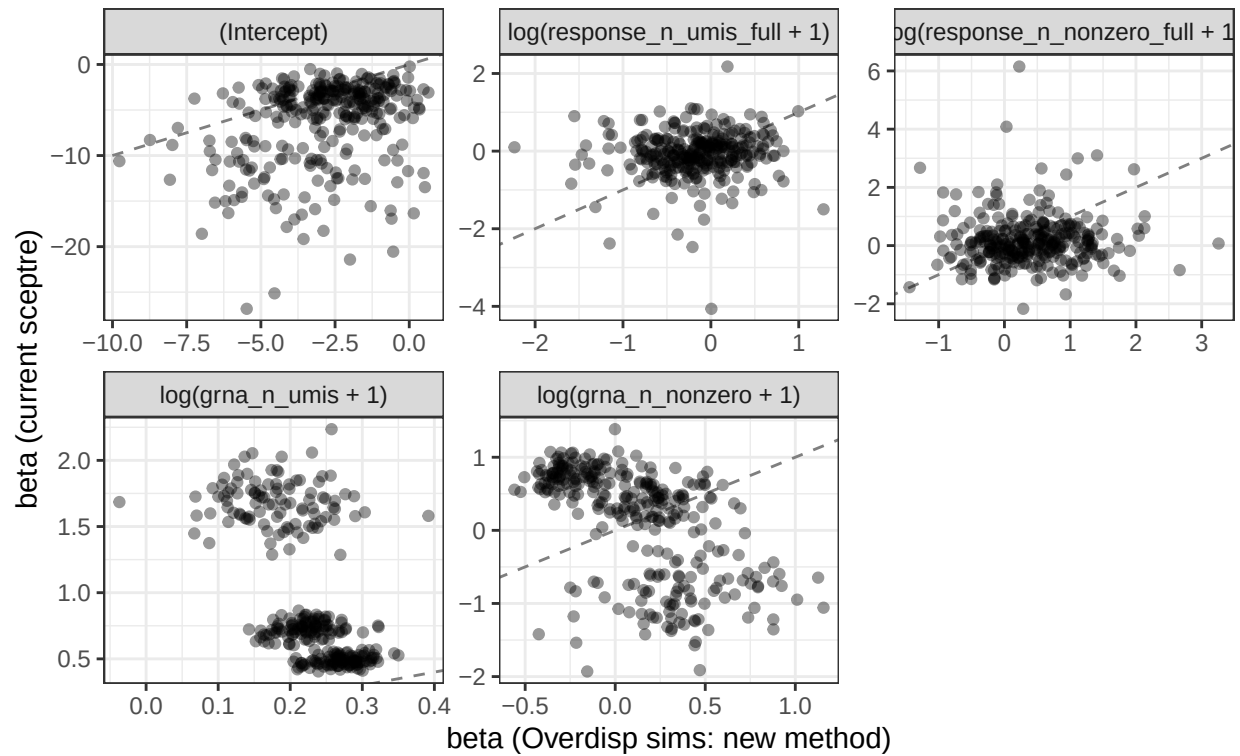
2. offset exploration

how do my different thresholdings work?

Per-guide offset-GLM coefficients: 2np_3_p sims: new method vs current
600 guides; 5 shared coefficient(s); facet = coefficient



Per-guide offset-GLM coefficients: Overdisp sims: new method vs current
 300 guides; 5 shared coefficient(s); facet = coefficient



The betas vary a lot, but that doesn't seem to be a huge performance effect. Maybe because bugged is robust to them?